AMR-CSI: Adaptive Multimodal RAG for Cold Start Indexing

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Abstract

Multimodal Retrieval-Augmented Generation (MM-RAG) systems enhance Large Language Models by grounding them in diverse knowledge sources. A key challenge for their practical deployment is the cold start problem: selecting an optimal indexing strategy (e.g., visual vs. textual) for new, unanalyzed document collections. A suboptimal choice severely degrades retrieval performance, while common workarounds like converting visual content to text are inherently lossy, destroying crucial information encoded in page layouts. This paper introduces Adaptive Multimodal RAG for Cold Start Indexing (AMR-CSI), a framework that addresses this problem by shifting from brute-force fusion to intelligent, upfront adaptation. AMR-CSI employs an Indexing Decision Router, a lightweight component that uses stochastic sampling to efficiently quantify a collection's visual density. Based on this analysis, the router automatically selects a single, optimal, modality-native indexing pipeline for the entire collection, eliminating the need for more complex architectures. Our empirical evaluation on heterogeneous corpora first provides quantitative evidence for the clear performance trade-off between specialized textual and visual indexing strategies. We then demonstrate that AMR-CSI achieves retrieval accuracy that closes the performance gap to a resource-intensive, fine-grained hybrid oracle to less than 2 percentage points, while inheriting the low latency and architectural simplicity of standard fixed-modality pipelines. By transforming the critical choice of indexing modality from a manual, error-prone decision into an automated, data-driven process, AMR-CSI provides a practical and scalable solution, removing a key architectural bottleneck that currently impedes the widespread adoption of robust, efficient, and factually grounded MM-RAG systems.

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CCS Concepts

• **Information systems** → **Information retrieval**; Information storage systems; • **Computing methodologies** → **Natural language processing**; *Computer vision.*

Keywords

Multimodal Retrieval-Augmented Generation, Adaptive Indexing, Information Retrieval, Visual Document Understanding, Large Language Models

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1 Introduction

Modern information environments are inherently multimodal, meaning today's knowledge bases frequently integrate heterogeneous data formats, including text, complex imagery, and structured tables. While Large Language Models (LLMs) have demonstrated impressive capabilities, their use in critical applications depends on Retrieval-Augmented Generation (RAG) [12]. By grounding LLM outputs in external, verifiable knowledge, RAG architectures enhance factual accuracy and reduce hallucination [4, 20].

Current RAG systems, however, are limited by a text-centric design that restricts their ability to interpret rich, non-textual information. This limitation leads to responses that can lack important context derived from visuals or layout. To overcome this, the field is advancing towards Multimodal Retrieval-Augmented Generation (MM-RAG), a paradigm that extends retrieval to a spectrum of data types [1]. The goal is to address documented LLM deficiencies by leveraging a more complete view of source knowledge. However, achieving this potential reveals fundamental challenges in how we index and retrieve from these complex sources.

The central difficulty lies in efficiently accessing relevant information from vast, heterogeneous knowledge bases. The conversion of non-textual data to a textual format, a conventional preprocessing step involving techniques like OCR or automated captioning, is inherently an information-degrading process that compromises semantic accuracy. A more fundamental challenge, however, arises before any retrieval operation: the architectural "cold start" problem. For any new, large-scale document collection, a developer must

make a crucial a priori decision on the primary indexing strategy (e.g., visual vs. textual). This choice is critical, as manual analysis is intractable at scale and a poor decision can severely degrade the performance of the entire system.

This paper directly addresses this foundational 'cold start' problem. We introduce Adaptive Multimodal RAG for Cold Start Indexing (AMR-CSI), a novel framework that automates the selection of an optimal, modality-native indexing strategy for entire document collections. Our approach is based on the observation of a clear performance trade-off: dense visual embeddings are superior for image- and layout-rich documents, while traditional textual embeddings excel for text-dense content. AMR-CSI makes a principled, data-driven decision upfront, thus eliminating the need for lossy cross-modal conversions and ensuring that each collection is indexed using its most effective native representation. This paper makes the following primary contributions: (1) A novel framework, AMR-CSI, that resolves the critical cold start problem by automatically selecting an optimal, modality-native (visual vs. textual) indexing strategy for entire collections, (2) A modality-exclusive indexing architecture that eliminates the information loss and system overhead inherent in conversion-based or hybrid MM-RAG systems, and (3) An extensive empirical evaluation demonstrating that the proposed adaptive approach achieves significant performance improvements on heterogeneous corpora compared to fixed-strategy baselines.

The rest of this paper is organized as follows. Section 2 reviews related work in retrieval-augmented generation and multimodal representation learning. Section 3 details the architecture and methodology of the AMR-CSI framework. Section 4 presents our experimental setup. Section 5 details our results and analysis. Finally, Section 6, 7 offers concluding remarks and outlines directions for future research.

2 Related Works

Early Retrieval-Augmented Generation (RAG) systems were developed to ground Large Language Model (LLM) outputs in external text, a proven technique for improving factual accuracy and mitigating hallucinations [4, 20]. However, the extension of this paradigm to multimodal documents has historically depended on converting all content into a singular text-only format. This common practice, which involves a pipeline of Optical Character Recognition (OCR) and automated captioning, is inherently lossy and introduces a significant performance bottleneck.

Recent studies confirm that errors originating from OCR directly degrade the quality of downstream retrieval and generation, particularly for documents with complex layouts [26]. Even advanced vision-language models (VLMs) produce incomplete or erroneous text when tasked with extraction from visually dense pages [9]. Following extraction, the text chunking process further disrupts the document's semantic and structural integrity [3], an added complexity that often fails to yield corresponding performance improvements [17]. This entire conversion process discards vital spatial information, disassociating text from its visual context and limiting the effectiveness of text-only retrievers such as MiniLM [24] or BGE-M3 [6].

To overcome these deficiencies, a paradigm shift toward vision-centric retrieval has gained momentum. This approach is enabled by powerful VLMs like CLIP [18] and BLIP [13], which learn shared embedding spaces for direct cross-modal retrieval. Systems such as VisRAG [25] and DSE [15] exemplify this method by embedding entire document pages as images, thereby preserving the original layout and visual context. Supporting research has demonstrated that native image embeddings consistently outperform generated text captions for retrieval tasks [5, 19]. While more sophisticated architectures like ColPali [8] have been introduced, their evaluations remain limited to image-heavy corpora. Recognizing that no single modality is universally optimal, hybrid systems have also emerged, which maintain parallel indexing pipelines for both text and images [2, 16]. These dual-stream systems, however, impose substantial computational overhead and deployment complexity [21].

The progress in vision-centric and hybrid RAG has exposed a crucial and unaddressed challenge. Although benchmarks like MMDocRAG [7] exist to evaluate retrieval performance, the literature lacks systematic guidance on selecting an optimal retrieval strategy for a given collection. For the large, uncharacterized document sets common in enterprise environments, this creates a significant "cold start" problem. An organization must either bear the high cost of a hybrid system or make a suboptimal, one-size-fits-all indexing choice that compromises performance. To our knowledge, no prior work has proposed a data-driven method to automatically determine the most effective native indexing strategy (i.e., visual vs. textual) for an entire collection at the point of ingestion.

Our work, AMR-CSI, directly addresses this critical gap. Rather than building a resource-intensive parallel system, we introduce a lightweight, automated framework that analyzes a collection to inform a one-time, strategic decision on the optimal indexing pipeline. By solving the cold start indexing problem, our approach ensures the knowledge base is represented in its most appropriate modality, thereby maximizing retrieval effectiveness while minimizing information loss and system complexity.

3 Methodology

The AMR-CSI architecture performs this strategic selection using three key components: (1) the Indexing Decision Router, which executes a lightweight pre-analysis of the collection to select the optimal indexing mode; (2) the Unified Indexer, which then processes the entire collection exclusively in the chosen mode; and (3) the Retrieval and Generation Module, which leverages the resulting optimized index to answer user queries, as shown in Figure 1.

3.1 Indexing Decision Router

The central component of the AMR-CSI framework is the Indexing Decision Router, designed to efficiently classify a large document collection D without manual analysis. The process begins with stochastic sampling of the collection to draw k representative subsets of pages, $\{S_1, S_2, ..., S_k\}$. For each sampled page p, a pre-trained layout detection model, PubLayNet [27] is employed to identify the set of textual bounding boxes, $B_{text}(p)$, and non-textual bounding boxes, $B_{non-text}(p)$ (e.g., figures, tables, images).

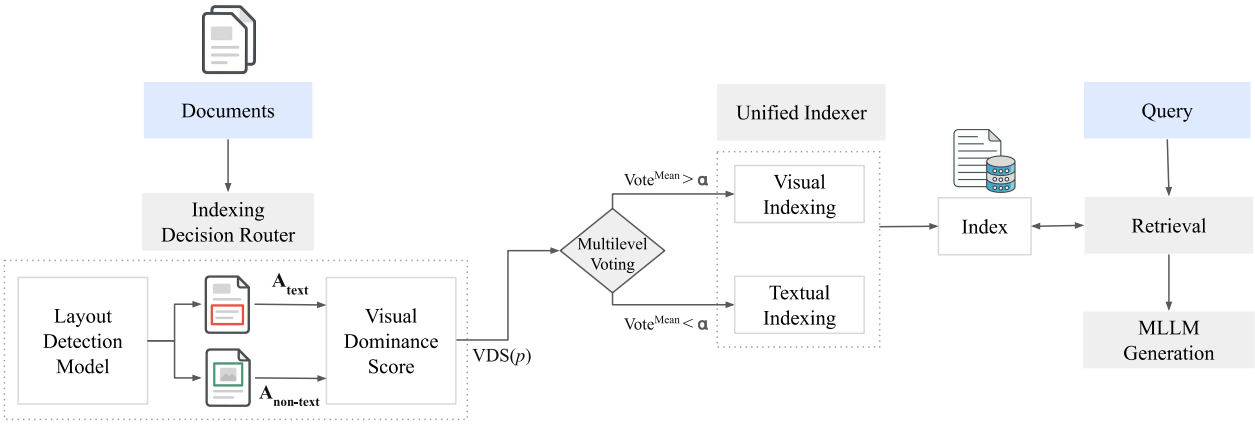


Figure 1: An overview of the AMR-CSI framework, highlighting its two-stage process: a one-time adaptive indexing pipeline and a standard query-time retrieval pipeline. The Indexing Decision Router (left) automates the solution to the 'cold start' problem. It first analyzes document layouts via a layout detection model to compute a page-level Visual Dominance Score (VDS). A multilevel voting mechanism then aggregates these scores into a statistically robust, collection-wide verdict. This verdict directs the Unified Indexer to operate in one of two mutually exclusive modes, creating either a purely visual or purely textual index for the entire collection. The resulting single, optimized index enables the Retrieval and Generation Module (right) to perform efficient query-time lookups without the overhead of merging multiple indexes, before passing the context to an MMLM.

The total area for each category is calculated by summing the areas of their respective bounding boxes:

$$A_{\text{text}}(p) = \sum_{b \in B_{\text{text}}(p)} \text{Area}(b) \tag{1}$$

$$A_{\text{non-text}}(p) = \sum_{b \in B_{\text{non-text}}(p)} \text{Area}(b) \tag{2}$$

where b denotes an individual bounding box and $Area(b)$ is its geometric area. These measurements are used to compute a normalized Visual Dominance Score (VDS) for each page, which quantifies the proportional area occupied by non-textual content:

$$\text{VDS}(p) = \frac{A_{\text{non-text}}(p)}{A_{\text{text}}(p) + A_{\text{non-text}}(p)} \tag{3}$$

A page-level vote is then cast based on its VDS relative to an empirically determined threshold τ, which represents the point at which non-textual content becomes semantically dominant:

$$\text{Vote}(p) = \begin{cases} 1, & \text{if VDS}(p) > \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where 1 represents a vote for Visual and 0 for Textual. The final verdict for the entire collection, Mode, is determined by aggregating votes across the sampled subsets to ensure a statistically robust decision:

$$\text{Mode} = \begin{cases} \text{Visual}, & \text{if } \frac{1}{k}\sum_{i=1}^{k} \text{Vote}(S_i) > \alpha \\ \text{Textual}, & \text{otherwise} \end{cases} \tag{5}$$

where $Vote(S_i)$ is the majority vote outcome for the i-th subset, and generally α is set to 0.5. This multi-level voting provides a robust, statistically grounded directive for the indexing pipeline.

3.2 Unified Indexer

Guided by the verdict from the router, the Unified Indexer operates in one of two mutually exclusive modes to process the entire collection. If the verdict is Visual, each page p is processed by an optimized ColPali pipeline, denoted E_{vis}, to generate a high-fidelity visual embedding $v_p = E_{vis}(p)$. Conversely, if the verdict is Textual, text is extracted from each document, segmented into a set of chunks $\{c_1, ..., c_m\}$, and each chunk c_i is embedded using a sentence-transformer model E_{txt} to produce $v_{c_i} = E_{txt}(c_i)$. The resulting set of vectors v is then stored in a single, unified vector index I.

3.3 Retrieval and Generation Module

During inference, a user query q is embedded using the same encoder (E_{vis} *or* E_{txt}) chosen during indexing to produce a query vector v_q. The system then performs a similarity search to retrieve the indices of the top-k most relevant items from the vector index I. This operation is defined as:

$$C_{\text{idx}} = \underset{j \in I}{\arg,\text{top-k}}\, \text{sim}(v_q, v_j) \tag{6}$$

where $sim(v_q <, v_j)$ is a similarity function such as cosine similarity, and the arg top-k operator returns the indices j corresponding

to the k highest similarity scores between the query vector v_q and the index vectors v_j.

The full-page images and their corresponding raw text for the retrieved indices constitute the multimodal context. This curated context is then passed to a Multimodal Large Language Model (MMLM) to synthesize the final, grounded answer.

4 Experiments

The central hypothesis motivating this work is that no single indexing modality is universally optimal for multimodal documents, necessitating an adaptive approach. The experimental evaluation is therefore designed to empirically validate this premise by examining critical performance trade-offs. Specifically, the investigation is structured to address three core questions: (i) How do text-centric and vision-centric retrieval strategies perform across document collections with varying modal compositions? (ii) Can modern page-level visual models be deployed scalably without prohibitive cost? (iii) How does the proposed collection-level adaptive strategy in AMR-CSI compare against both fixed and computationally intensive hybrid approaches?

4.1 Experimental Setup

To systematically address these questions, this study employs a rigorous experimental design comprising carefully selected datasets, a comprehensive suite of evaluation metrics, and a purposefully chosen set of comparative strategies.

4.2 Datasets

The foundation of the evaluation rests on a diverse corpus, detailed in Table 1, which was curated to simulate the heterogeneous collections found in real-world enterprise environments. This corpus deliberately includes both purely textual datasets (e.g., 2WikiMultihopQA [10], MultiHopRAG [22]) and inherently multimodal datasets (the Vidore suite [8]), creating a controlled setting essential for answering RQ1 by precisely measuring the performance trade-offs of different modalities. A crucial methodological step is to render all documents as PDF pages, which could subsequently be processed as images. This pipeline ensures that vision-native models can be applied across the entire corpus, creating a level playing field for a truly modality-agnostic comparison and avoiding any implementation bias.

4.3 Evaluation Metrics

Performance is assessed using a dual-focus protocol that reflects the practical constraints of real-world deployment. System effectiveness is measured through Recall@k (k=5, 10), which quantifies the accuracy of the retrieval subsystem, and the final accuracy of answers from the downstream MMLM, which assesses end-to-end quality. To address RQ2 and quantify the practical deployment costs central to RQ3, efficiency is also measured via total Indexing Time and average Query Latency.

4.4 Comparative Strategy Evaluation

The core of the evaluation is a direct comparison of four distinct indexing strategies, each representing a key philosophy in system design. The comparison begins with two Fixed Baselines: a Fixed-Textual and a Fixed-Visual architecture, each applied uniformly across all datasets. These represent the conventional, one-size-fits-all approach and serve to establish the performance boundaries of non-adaptive systems. To establish a practical performance ceiling, a Fine-Grained Hybrid baseline is also implemented. This resource-intensive model analyzes every page individually, maintains two separate indexes, and merges results at query time. While expected to be highly accurate, this strategy represents a "brute-force" solution whose significant overhead serves as a key point of comparison for efficiency. Positioned against these alternatives is the proposed AMR-CSI framework. It employs its Indexing Decision Router to make a single, data-driven, collection-level decision for each dataset. The guiding hypothesis for RQ3 is that AMR-CSI will achieve retrieval accuracy comparable to the far more complex Fine-Grained Hybrid baseline while maintaining an efficiency profile closer to the simpler Fixed strategies, thereby demonstrating a superior performance-to-cost trade-off.

5 Results and Analysis

The empirical results validate the central hypothesis of this work, demonstrating a clear need for adaptive indexing and confirming the efficacy of the proposed AMR-CSI framework. The analysis begins by establishing the fundamental performance trade-offs that necessitate an adaptive approach.

Initial experiments comparing fixed retrieval strategies reveal a clear performance divergence, as shown in Table 2. The Text-Centric strategy, which relies on linearized document content, naturally achieves a high recall of 0.812 on the text-dense MultiHopRAG dataset. However, its effectiveness collapses on visually complex documents, with performance dropping by over 15 percentage points on the Vidore datasets where crucial layout and figure context is lost. Conversely, the Page-Level Visual model, which processes documents as holistic images, dominates on multimodal data. It outperforms the text-centric approach on the Vidore-Government dataset by over 26%, confirming its superior ability to interpret complex layouts. This empirically confirms that a fixed, one-size-fits-all indexing strategy is inherently suboptimal, as it guarantees significant underperformance on documents not aligned with its chosen modality.

A prerequisite for an adaptive system that includes a visual option is the practical scalability of its visual component. As detailed in Table 3, a baseline brute-force visual retrieval is prohibitively slow, with a latency of 5,230 ms that makes it unsuitable for real-time applications. However, standard optimizations prove highly effective at mitigating this concern. A phased retrieval strategy, which combines a fast initial search using a compact HNSW index with a precise re-ranking step [23], reduces this latency by nearly 95% to a practical 261 ms. This approach also shrinks the memory footprint by over 12x, from 2.3 GB to just 183 MB, with negligible impact on recall. These findings are critical, as they confirm that page-level visual indexing is a viable and scalable strategy that does not pose a prohibitive computational barrier to large-scale deployment.

Table 1: Overview of the heterogeneous corpus designed for the experimental evaluation. The datasets were selected to create a controlled environment essential for validating the paper's central hypothesis. The collection includes two text-dense benchmarks and three visually-rich public report datasets.

Dataset	Type	Domain	# Contexts	# Questions
2WikiMultihopQA	Textual	General Knowledge	5700	570
MultiHopRAG	Textual	General Knowledge	762	660
Vidore-Government	Multimodal	Government Reports	980	100
Vidore-Health	Multimodal	Health & Medical	980	100
Vidore-Energy	Multimodal	Energy Sector	980	100
Total	-	-	**9402**	**1530**

Table 2: Empirical validation of the central performance trade-off between fixed indexing strategies. This table presents the Recall@10 for Text-Centric (GTE [14]), Shared-Space (Jina-CLIP [11]), and Page-Level Visual models across the heterogeneous corpus. The results demonstrate a clear pattern: the text-native model achieves the highest scores on text-dense datasets, whereas the vision-native model (ColPali) dominates on the multimodal Vidore datasets where layout is critical. The highest performance for each dataset is highlighted in bold.

Retrieval Strategy	2WikiMultihopQA	MultiHopRAG	Vidore-Gov.	Vidore-Health	Vidore-Energy
Text-Centric (GTE)	**0.783**	**0.812**	0.651	0.633	0.598
Shared-Space (Jina-CLIP)	0.705	0.734	0.720	0.745	0.711
Page-Level Visual (ColPali)	0.698	0.715	**0.825**	**0.817**	**0.758**

Table 3: Analysis of scalability optimizations for the Page-Level Visual model, a core component of the AMR-CSI framework. This experiment evaluates the impact of standard techniques on the model's high baseline latency and memory footprint. The results show that while a brute-force approach is impractical (5230 ms), a Phased retrieval strategy combining HNSW indexing with embedding quantization achieves the optimal trade-off. It reduces latency by nearly 95% to 261 ms and shrinks the index size by over 12x, all with a negligible impact on recall.

Method (10k Samples)	Recall@5 (Avg. Textual)	Recall@5 (Avg. Multimodal)	Latency (ms)	Index Size
Brute Force (float-float)	0.435	0.973	5230	2.3 GB
HNSW (float-float)	0.535	0.978	505	2.3 GB
HNSW (float-binary)	0.499	0.980	613	183 MB
HNSW (Phased)	0.476	0.960	261	183 MB

Table 4: Overall comparison of indexing strategies across the full heterogeneous corpus, evaluating the trade-off between retrieval accuracy (Recall@10), speed (Avg. Query Latency), and complexity. The Fixed strategies serve as efficiency benchmarks, while the Fine-Grained Hybrid acts as a resource-intensive performance oracle. The proposed AMR-CSI framework, highlighted in bold, achieves aggregate recall that closes the gap to the oracle to less than 2 percentage points, while operating at a query latency comparable to the far simpler Fixed models.

Strategy	Aggregate Recall@10	Avg. Query Latency (ms)	Indexing Complexity
Fixed-Textual	0.700	245	Low (1 Index)
Fixed-Visual	0.743	261	Low (1 Index)
Fine-Grained Hybrid	0.815	550+	High (2 Indexes + Re-ranker)
AMR-CSI (Ours)	**0.801**	**253**	**Low (1 Index + Router)**

With the need for adaptation established and the viability of visual indexing confirmed, the definitive evaluation compares AMR-CSI against the fixed and fine-grained hybrid baselines. The results in Table 4 demonstrate AMR-CSI's superior balance of accuracy and efficiency. The expensive Fine-Grained Hybrid model, which maintains dual indexes and performs complex query-time merging, serves as a performance oracle. It achieves the highest aggregate recall of 0.815 but pays a prohibitive cost, requiring over 550 ms of

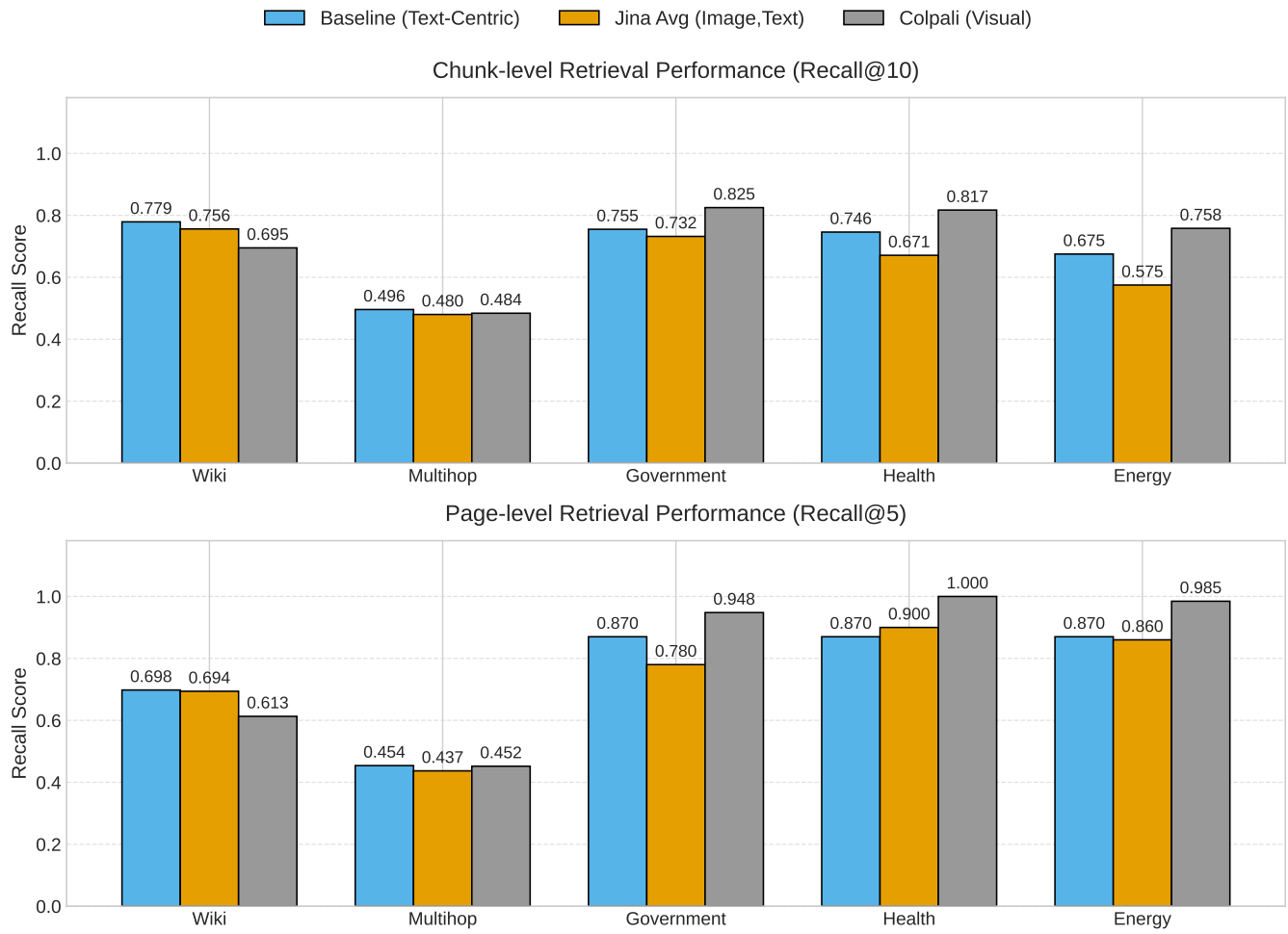


Figure 2: Analysis of retrieval performance across chunk-level (top) and page-level (bottom) granularities to evaluate the interplay between data modality and indexing strategy. The experiment compares Text-Centric, Shared-Space, and Page-Level Visual models. The results reveal that the performance of the visual model (Colpali) is substantially amplified when operating at its native page-level granularity on multimodal documents. Its recall on Vidore-Health reaches a near-perfect 1.0 at the page level. This provides strong evidence that preserving the native, holistic representation of a document is paramount for optimal performance on visually complex data.

query latency. By contrast, AMR-CSI achieves an aggregate recall of 0.801, closing the performance gap to the oracle model to less than 2%, while maintaining a query latency of only 253 ms. This makes it more than twice as fast as the hybrid approach and on par with the simple fixed models. This result powerfully demonstrates that the framework's strategic, data-driven decision at index-time delivers near-optimal accuracy while entirely avoiding the prohibitive architectural complexity, storage overhead, and query latency of a full hybrid system.

These conclusions are reinforced by further analysis into retrieval granularity and performance in a simulated real-world environment. Analyzing granularity reveals that the visual model's advantage is amplified when operating on its native page-level data. For instance, on the Vidore-Health dataset (Figure 2), the visual model's recall leaps from 0.817 at the chunk level to a near-perfect 1.0 at the page level, demonstrating the importance of processing the holistic visual layout. This evidence culminates in the unified index stress test, which magnifies the failure modes of non-adaptive systems. In such a realistic mixed environment, a Fixed-Textual model would be unreliable, with its recall dropping to as low as 0.598 for the visually-driven Vidore documents. Similarly, a Fixed-Visual retriever would be equally ineffective on the text-dense corpora. This predictable, modality-dependent failure of fixed strategies in a mixed setting provides the definitive justification for AMR-CSI, confirming its collection-level analysis is essential for ensuring robust and reliable performance across an entire, diverse knowledge base.

6 Discussion

The empirical results from this study challenge the prevailing architectural trend in MM-RAG, which often favors increasingly complex, fine-grained hybrid models. The prevailing premise is that comprehensive fusion of all available modalities is always superior. These results suggest a more nuanced reality. While the resource-intensive Fine-Grained Hybrid baseline did achieve the highest aggregate recall, this marginal gain ($\sim 2\%$) came at a prohibitive cost: more than double the query latency and a significant increase in system complexity compared to AMR-CSI. By contrast, AMR-CSI delivered near-oracle accuracy with the efficiency and architectural simplicity of a standard fixed pipeline. This work therefore posits that the objective for many large-scale applications should not be to build systems that fuse all modalities by default, but rather to design systems that intelligently select the optimal modality for a given data collection, maximizing the return on computational investment.

Furthermore, the success of the Page-Level Visual strategy on specific datasets provides a clear technical justification for this adaptive approach. The superior performance of the ColPali model on multimodal documents underscores that critical information is encoded not just in text, but in its two-dimensional layout, structure, and proximity to visual elements. Conventional methods that rely on lossy conversions to text fundamentally discard this spatial context. The ability of a native visual retriever to process a page as a holistic unit unlocks a deeper level of semantic understanding that text-based representations cannot capture. This highlights a critical direction for future MM-RAG development: a shift away from improving intermediate textual representations and towards advancing retrievers that operate on modalities in their native form.

While the findings are robust, this study has limitations that present clear avenues for future research. First, the Indexing Decision Router in its current form makes a binary, collection-wide choice between a purely textual or purely visual pipeline. For document collections that are truly and uniformly mixed within documents, a more directed hybrid approach might still be superior. Future iterations could extend the router's decision space to include "Hybrid" as a third potential verdict, to be chosen only when the variance of visual content across the collection is high. Second, while the area-based visual density score proved effective, it could be enhanced with more sophisticated, learnable layout analysis features to better distinguish between decorative images and information-rich charts or diagrams, further refining the router's decision-making capabilities.

7 Conclusion

This work addresses the cold start problem: a fundamental challenge in deploying Multimodal Retrieval-Augmented Generation systems on large, uncharacterized document collections. The study first establishes that fixed, one-size-fits-all indexing strategies are inherently suboptimal, as their performance is highly dependent on the dominant modality of the underlying data.

To resolve this issue, this paper introduces Adaptive Multimodal RAG for Cold Start Indexing (AMR-CSI), a novel framework featuring an Indexing Decision Router. This component employs efficient stochastic sampling and layout analysis to automate the selection of the most effective indexing pipeline—either visual or textual—for an entire collection. The empirical evaluation confirmed a clear performance trade-off between these specialized strategies and demonstrated that AMR-CSI achieves retrieval accuracy nearly on par with a resource-intensive, fine-grained hybrid system, but at a fraction of the associated latency and deployment complexity.

The principal contribution of AMR-CSI lies in transforming the strategic choice of modality from a manual, ad-hoc decision into an automated, data-driven process at the point of ingest. This provides a practical, scalable, and high-performance solution that streamlines the MM-RAG pipeline and minimizes information loss. Ultimately, this framework enables the development of more robust, contextually aware, and factually grounded AI systems for real-world enterprise applications.

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